

Assignment 05

Third Year BS(Honours) 2024-2025

Course Title: Math Lab III Course Code: AMTH 350

Department of Applied Mathematics, University of Dhaka

Name:

Roll:

Group:

1. Testing for convexity using the graphical method for the following sets:

- i) $R = \{(x, y): x^2 + y^2 \leq 1\}$
- ii) $R = \{(x, y, z): 2x - y + 3z \leq 2\}$
- iii) $R = \{(x, y): xy \geq 1, x, y \geq 0\}$
- iv) $R = \{(x, y): y \leq x^2\}$

2. One kind of cake requires 200 g of flour and 25g of fat, and another kind of cake requires 100 g of flour and 50 g of fat Find the maximum number of cakes using Graphical Method that can be made from 5 kg of flour and 1 kg of fat assuming that there is no shortage of the other ingredients, used in making the cakes.

3. A nutritionist wants to minimize the cost of a diet plan while meeting minimum requirements for vitamins and minerals. Food 1 costs Tk.5/kg, and Food 2 costs Tk. 6/kg. Need at least 90 units of Vitamin X and 120 units of Mineral Y.

Food 1: Contains 4 units of Vitamin X and 5 units of Mineral Y per kg.

Food 2: Contains 7 units of Vitamin X and 4 units of Mineral Y per kg.

Formulate the above LPP and solve it using graphically method.

4. The supply (from Company) and the demand (from Retail) together with the unit transportation costs per Shipping on the different routes are summarized in the transportation model, given below:

Company	Retail				Supply
	R1	R2	R3	R4	
P1	3	5	7	6	50
P2	2	5	8	2	75
P3	3	6	9	2	25
Demand	20	20	50	60	

Write a MatLab code to find the initial basic feasible solution using :

- North-West Corner Method
 - Least Cost Method
5. Write a MatLab program to find the optimal solution of the transportation problem describe in Q.(4) using MODI Method.
6. Write down a MATLAB code to find the value of the game of the following game:

$$\begin{pmatrix} 2 & 1 & 2 \\ 1 & -5 & -4 \\ 2 & 6 & -2 \end{pmatrix}$$